

Model calibration

**KNOWLEDGE DISTILLATION (KD) -
an effective method to transfer
knowledge from one network (the
teacher) to another one (the student).**

MODEL CALIBRATION

PART I

TEACER MODEL CALIBRATION

(THE CASE OF STANCE DETECTION)

➤ It has been shown that:

- transferring knowledge from a teacher model with desired high performance to a student model, through knowledge distillation (KD), can lead to better performance of student models distilled (Furlanello et al., 2018; Yim et al., 2017);
- calibration measures the discrepancy between the correctness of the prediction (i.e. accuracy) and the (empirical) probability that a model assigns to a prediction (i.e. confidence).
- a well-calibrated model *knows* how often it is correct or wrong; predicting an event with p confidence shall empirically be true p of the time (Guo et al., 2017).

NOTATIONS:

KD $\equiv$ Knowledge Distillation

AKD $\equiv$ Adaptive Knowledge Distillation

**CKD $\equiv$ Calibration – based Knowledge
Distillation**

ECE $\equiv$ Expected Calibration Error

Teacher Calibration for Stance Detection

(ACL 2023)

- **Li and Caragea (2023) study the problem of knowledge distillation in generations on stance detection. The paper:**

Li, Y. and Caragea, C., Distilling Calibrated Knowledge for Stance Detection. ACL 2023, July 9-14, 2023, pp. 6316 -6329.

➤ The problem in stance detection: teacher models of AKD trained on stance detection datasets are not well-calibrated, producing peaked distributions of confidence.

➤ The suggested solution (Li and Caragea, 2023):

- to associate the performance of knowledge distillation with the calibration of teacher models;
- to further improve the task performance by calibrating teacher predictions.

Knowledge Distillation (KD)

- Knowledge Distillation (KD) (Hinton et al., 2015) transfers knowledge from a teacher model to a student model by training the student model to imitate the teacher's prediction logits (which are called *soft labels*).
- Note that:
 - ✓ In knowledge distillation, a student model is trained based on two signals: the *hard labels* and the teacher estimates, which are known as *soft labels*.
 - ✓ A *hard label* is a label assigned to a member of a class where membership is *binary*; a *soft label* is one which has a *score* (probability or likelihood) attached to it.

- **Model miscalibration can be widely observed in modern neural networks (Guo et al., 2017; Yang and Song, 2021; Guo et al., 2021).**
- **Poorly calibrated model $\equiv$ is over-confident in its predictions**
- **If a model is well calibrated, then the probability associated with the predicted label should reflect its ground-truth correctness.**

➤ Here:

- Li and Caragea (2023) recalibrate teacher predictions using a post-processing method called temperature scaling (Platt, 1999; Guo et al., 2017).
- The student model is trained based on both hard labels and calibrated teacher predictions.

HERE (for stance detection):

- Li and Caragea (2023) use multiple student generations.
- Li and Caragea (2023) explore the born-again networks for stance detection and propose to calibrate teacher predictions in each generation.
- Born-again networks (Furlanello et al., 2018): the teacher and student models have identical model architectures; at each consecutive step, a student model is taken as a new teacher to transfer knowledge to a new fresh student model.

THE APPROACH

1. Knowledge Distillation

- Suppose a given stance dataset of size n is

$$D^{tr} = \{(x_i, y_i)\}_{i=1}^n$$

where

$x_i \equiv$ includes an input sentence and a corresponding target

$y_i \equiv$ the hard label

- The goal of stance detection is to predict the stance label given an input sentence and a target.
- Standard supervised learning aims to minimize the cross-entropy loss of training data:

$$L_{CE} = \sum_{(x_i, y_i) \in D^{tr}} l_{CE}(p(x_i), y_i) \quad (1)$$

where:

$l_{CE}(\cdot) \equiv$ cross entropy

$p(x_i) = \sigma(z(x_i)) \equiv$ softmax predictions of the model.

- **Knowledge distillation** (Hinton et al., 2015) transfers knowledge from a teacher model to a student model by minimizing L_{KD} , which is:
 - ✓ the sum of the cross-entropy loss between the student predictions and hard labels and the distance loss between the predictions of the student and those of the teacher (see formulae (2) and (3) on the next slide).

$$L_{KD} = (1 - \lambda)L_{CE} + \lambda L_{KL} \quad (2)$$

$$L_{KL} = \sum_{x_i \in D^{tr}} l_{KL}(p_s^\tau(x_i), p_t^\tau(x_i)) \quad (3)$$

where

$L_{KL} \equiv$ the Kullback-Leibler (KL) divergence loss

$p_t^\tau(x) = \sigma(z_t(x)/\tau) \equiv$ the softmax output of the teacher model

$p_s^\tau(x) = \sigma(z_s(x)/\tau) \equiv$ the softmax output of the student model

with temperature scaling τ , where:

$\tau \equiv$ the fixed temperature used to scale the model predictions

$\sigma(\cdot) \equiv$ the softmax function

$z_t(x) \equiv$ the output logit of the teacher model

$z_s(x) \equiv$ the output logit of the student model

$\lambda \equiv$ a coefficient to balance the importance of the two loss functions

2. Calibration of Neural Networks

- Miscalibration can be widely observed in modern neural networks.
- One notion of miscalibration is Expected Calibration Error (ECE) (Naeini et al., 2015).
 - *ECE* represents the difference in expectation between confidence and accuracy:

$$ECE = \mathbb{E}_{\hat{P}} \left[\left| \mathbb{P}(\hat{Y} = Y / \hat{P} = p) - p \right| \right] \quad (4)$$

where:

$Y \equiv$ the ground-truth label

$\hat{Y} \equiv$ the predicted label

$\hat{P} \equiv$ the output probability associated with the predicted label

$p \in (0, 1) \equiv$ a given confidence

- As Eq. (4) cannot be estimated using a finite number of samples if $\hat{P}$ is a continuous random variable, empirical approximations (Guo et al., 2017) are usually adopted by partitioning predictions into M bins of equal size and computing the weighted average as in (5),

$$ECE = \sum_{m=1}^M \frac{|B_m|}{n} |acc(B_m) - conf(B_m)| \quad (5)$$

where:

$B_m \equiv$ the set of samples whose prediction confidence falls into the interval $\left(\frac{m-1}{M}, \frac{m}{M}\right)$

$M \equiv$ the bin size

$acc(B_m) \equiv$ accuracy

$conf(B_m) \equiv$ average confidence of B_m

- **Temperature scaling** (Guo et al., 2017; Desai and Durrett, 2020) is an effective post-processing calibration method that produces calibrated probabilities.
- Given the logit vector $z(x)$, the new confidence prediction is $\sigma(z(x)/\tau)$ after temperature scaling with temperature τ being usually fixed in knowledge distillation.

3. Calibrated Knowledge Distillation (CKD)

- **Li and Caragea (2023) propose an improved knowledge distillation method called CKD (see Algorithm 1; next slide).**
- **It is shown (Li and Caragea, 2023) that transferring knowledge from a well-calibrated teacher improves the performance of students (for stance detection).**
- **The proposed CKD (Li and Caragea, 2023) produces calibrated teacher predictions by dynamically updating the temperature used for scaling in each generation (see Algorithm 1; next slide)**

Algorithm 1: Calibrated-Knowledge Distillation with Dynamic Temp. Scaling

Require: Train set $D^{tr} = \{(x_i, y_i)\}_{i=1}^n$
Val set D^{val}

1. Train the first teacher model by minimizing the cross-entropy loss of D^{tr}

$$L_{CE} = \sum_{x_i \in D^{tr}} l_{CE}(p(x_i), y_i)$$

2. Do temperature scaling of teacher predictions with updated τ of temperature scaling that minimizes the *ECE* of teacher predictions on the D^{val}
3. Train a student model by minimizing the sum of cross-entropy loss and the KL-divergence loss of D^{tr}

$$L_{KD} = (1 - \lambda)L_{CE} + \lambda L_{KL}$$

4. Iterative training: Use the student as new teacher and go back to step 2
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➤ Algorithm 1

- The idea:

In each consecutive step, a new student model is initialized with an identical model architecture and is trained based on both hard labels and calibrated teacher predictions.

- The rationale:

The calibrated teacher predictions are expected to help students learn better *inter-class similarity* and achieve better performance.

- The method:

The proposed CKD produces calibrated teacher predictions by dynamically updating the temperature used for scaling in each generation.

Experimental Settings

1. Datasets

- **AM** (Stab et al., 2018) is an argument mining dataset (argument mining $\equiv$ automatically identifying and extracting argumentative structures in discourse);
 - ✓ it contains 8 topics (abortion, cloning, death penalty, gun control, Marijuana legalization, minimum wage, nuclear energy, school uniforms);
 - ✓ the dataset is annotated for detecting whether an argument is in support of, neutral or opposed to a given topic.

- **COVID-19** (Glandt et al. 2021)
 - ✓ it contains 4 targets: “face mask”, “Fancy”, “stay at home orders” and “school closures”;
 - ✓ the dataset is annotated for detecting whether the author is in favor of, neutral or against these topics.
- **P-Stance** (Li et al. 2021) – collected during the 2020 U.S. presidential election
 - ✓ it contains 3 public figures: “Donald Trump”, “Joe Biden”, and “Bernie Sanders”;
 - ✓ the dataset is annotated for detecting whether the author is in favor of or against these presidential candidates.

2. Evaluation Metrics

- The performance of the models is evaluated with:
 - ✓ macro-average of F1-score (F_{macro})
 - ✓ micro-average of F1-score (F_{micro})
- We first calculate F_{avg} for each target
 - ✓ F_{avg} is calculated by averaging the F1-scores of label “Favor” and “Against”.
- F_{macro} is calculated by averaging the F_{avg} across all targets for each dataset.
- F_{micro} is calculated by averaging the F1-scores of “Favor” and “Against” across all targets for each dataset.

3. Baseline Methods

- **The strong baselines of stance detection are FNN, BiCE, TAN, CrossNet, BERT, TGA, BERT-KE, AKD, KD-1, KD-T, LSR, CKD for which see (Li and Caragea, 2023).**

4. Software

- **The model is implemented (Li and Caragea, 2023) in PyTorch using HuggingFace Transformers library.**
- **For details concerning the hyperparameters see Appendix A of (Li and Caragea, 2023).**

RESULTS

1. Main Result

(CKD $\equiv$ Calibration-based Knowledge Distillation
AKD $\equiv$ Adaptive Knowledge Distillation)

- The proposed CKD (Li and Caragea, 2023) performs best in overall. Specifically: the best student model of CKD outperforms the best-performing baseline AKD by
 - ✓ 1.86% in F_{micro} on COVID-19
 - ✓ 2.10% in F_{micro} on AM
 - ✓ 0.36% in F_{micro} on P-stance
- Remark: CKD shows less improvements on the P-stance dataset!
 - ✓ Explanation: the P-stance dataset has much larger train set for targets and the effect of knowledge distillation diminishes with increasing the size of the train set (Zhang and Sabuncu, 2020).

2.**Main Conclusion**

The proposed CKD shows the best performance, achieving the lowest ECE (Expected Calibration Error) on each dataset. The best student of the proposed CKD achieves the highest micro-averaged F1 on all datasets.

➤ Final Conclusion

A well-calibrated teacher can benefit the student by providing smoother training signals and makes it possible for the student to learn from inter-class similarity.

➤ Future Work

should include extending the proposed method to a broader range of NLP tasks (such as emotion classification).

MODEL CALIBRATION

PART II

STUDENT MODEL CALIBRATION

(THE CASE OF EMOTION DETECTION)

Calibrating Student Models for Emotion-related Tasks

- Hosseini and Caragea (2022) empirically examine the impact of the mixed sample data augmentation technique, MixUp (Zhang et al., 2018), on the performance and calibration of the student models in a distillation setup.
- MixUp (Zhang et al., 2018) is a data augmentation and regularization technique that generates a weighted combination of random input pairs from the training data.
- Hosseini and Caragea (2022) propose a simple mixup strategy to attain more accurate and better-calibrated models.

- **The paper:**

Hosseini, M., Caragea, C., Calibrating Student Models for Emotion-related Tasks. In: Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, December 7-11, 2022, pp. 9266 - 9278.

- **What it studies: using mixup for calibrating the student model predictions on the knowledge distillation setting.**

- **The main goal:** by employing MixUp, to provide the teacher models with more useful information and impart the student models with better supervision signals during the distillation of the emotion-related models.
- **Additionally:** MixUp may introduce some noise to the training data, which enables us to gain additional information about relatively similar data. This makes *teacher models more robust*, helping the *student models* to be *more accurate* and produce better-calibrated predictions.

- Hosseini and Caragea (2022) propose a simple mixup strategy, that is informed by training dynamics, for calibrating the student models. To contextualize examples in the datasets based on training dynamics, they use data maps.
- The carried out experiments create teacher networks on two pre-trained models, BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019).
- Hosseini and Caragea (2022) propose to mix easy-to-learn and ambiguous samples in the teacher model for the student model calibration.

METHODS

1. Teacher-Student Training

Given a k -class classification task and a dataset $D = \{x_i, y_i\}_{i=1}^n$ consisting of sentence-label pairs, standard supervised learning is optimized based on the one-hot labels by minimizing the cross-entropy loss $\mathcal{L}_{ce}$ of training data which is defined as:

$$\mathcal{L}_{ce}(p, y) = - \sum_{i=1}^n \sum_{j=1}^k y_{ij} \log p(y = j | x_i) \quad (1)$$

where p indicates the softmax outputs.

- In *knowledge distillation*, a *teacher-student training method* is employed to enhance the performance of the student model where the softmax outputs of the teacher model, p^t , is computed as:

$$p_i^t = \frac{\exp(z_i/\tau)}{\sum_j \exp(z_j/\tau)} \quad (2)$$

where:

$\tau \equiv$ the softmax temperature

$z \equiv$ the logits from the teacher model.

- In general, the knowledge distillation framework (Hinton et al., 2015) incorporates the knowledge obtained from the logits of a teacher model and transfers the knowledge to a small student model. In this way, better training signals can be retrieved from the data using a teacher-student framework.
- This is done by minimizing the sum of cross-entropy loss between hard labels and student's predictions and the difference loss between the student's and teacher's predictions:

$$\mathcal{L}_{KD} = \alpha \mathcal{L}_{ce}(\mathbf{p}, \mathbf{y}) + (1 - \alpha) \mathcal{L}_{KL}(\mathbf{p}, \mathbf{p}^t) \quad (3)$$

where:

$\mathcal{L}_{KL} \equiv$ is Kullback-Leibler (KL) divergence loss

$\alpha \in [0, 1] \equiv$ is the hyper-parameter that controls the impact of cross-entropy loss and the KL divergence loss.

2. MixUp Training

Given a training dataset of sentence-label pairs

$$D_{train} = \{(x_i, y_i)_{i=1}^n\}$$

and a language model f , standard MixUp creates the vicinal dataset by calculating a weighted average of training points based on the following simple rule (Zhang et al., 2018)

$$(\tilde{x}_{ij}, \tilde{y}_{ij}) := (\lambda x_i + (1 - \lambda)x_j, \lambda y_i + (1 - \lambda)y_j) \quad (4)$$

where:

(x_i, y_i) and $(x_j, y_j) \equiv$ two input examples that are randomly drawn from the training set

$\lambda \equiv$ a weight sampled from a beta distribution $\beta(\alpha, \alpha)$ with parameter $\alpha > 0$ taken to be relatively small

- **Standard MixUp (Zhang et al., 2018)** augments the training data by linearly interpolating training samples and their corresponding labels in the input space.
- **Park and Caragea (2022)** study random pairing (as in standard MixUp) versus informative pairing for MixUp. They propose a novel mixup data augmentation technique on the teacher models (*TD-MixUp*), that is informed by training dynamics, to improve the student model calibration on the distillation objective.
- **The proposed mixup creates vicinal distribution steered by the data maps (Swayamdipta et al., 2020) → see the previous lecture on dataset cartography.**

- **The paper introducing TD-MixUp**
(MixUp with training dynamics):

Seo Yeon Park and Cornelia Caragea, A Data Cartography based MixUp for Pre-trained Language Models. In: Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, July 10-15, 2022, pp. 4244 – 4250.

3. Calibration

- A probabilistic model is considered calibrated if its predicted probabilities of classes are equivalent to the actual probabilities of those classes. (Example: if a model allots 80% posterior probability to a class, that class should appear 80% of the time).
- To evaluate calibration it is best to use *the expected calibration error* – ECE (Naeini et al., 2015).

- **ECE measures model miscalibration by binning the predicted probabilities and measuring the gap between them and the average accuracies of these bins,**

$$\sum_{s=1}^S \frac{b_s}{N} |acc(s) - conf(s)|$$

where:

$S \equiv$ the overall number of bins

$b_s \equiv$ the number of predictions in the s -th bin

$N \equiv$ the total number of data points

$acc(s) \equiv$ the accuracy of the s -th bin

$conf(s) \equiv$ the confidence of the s -th bin

- **The presented experiments have used $S = 10$.**

EXPERIMENTS (Hosseini and Caragea, 2022)

1. Tasks and Datasets

- **Tasks:** evaluation is performed on three text classification tasks:
 - ✓ emotion detection
 - ✓ sentiment analysis
 - ✓ empathy detection

- **Datasets:**
 - ✓ Both in-domain and out-of-domain datasets are used: GoEmotions, Meld, Yelp, Stanford Sentiment Treebank (SST-2), EPITOME, NewsEmpathy
 - ✓ See (Hosseini and Caragea, 2022), pp.9271 for a description of these datasets.

2. Models

- Hosseini and Caragea (2022) perform the following study:
- to evaluate calibration of large-scale pre-trained models they fine-tune BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019)
 - they experiment with self-distillation (a particular case of the teacher-student training) where both teacher and student models have the same architecture (Zhang and Sabuncu, 2020). Note that: In this setting, a pre-trained language model like BERT or RoBERTa is first trained as a teacher model and then a student model with the same architecture is trained to mimic the output of the teacher model
 - they experiment with knowledge distillation, where knowledge from a pre-trained language model with a different architecture than the student model is distilled.

Model performance and calibration

The performance and calibration of the standalone and distilled models are compared using standard MixUp (Zhang et al., 2018) and the mixup method proposed in (Park and Caragea, 2022).

3. Results

- 1. Distillation leads to improved accuracy and model calibration compared to the standalone models (BERT or RoBERTa), both in the in-domain and out-of-domain settings.**
- 2. Compared to the vanilla pre-trained models (with or without distillation), label smoothing with temperature scaling does not always improve the calibration or accuracy of the models, specifically in the in-domain setting.**
- 3. No significant improvements in calibration or performance are observed by solely incorporating plain mixup (i.e. MixUp) on the standalone models (i.e. BERT or RoBERTa). Similarly, it is observed that if a teacher model is trained using plain mixup, the student model distilled from it is impaired in calibration and its generalization capabilities in most cases.**
- 4. When using informed mixup, the obtained errors are much smaller in general compared to the other settings.**

CONCLUSIONS

- 1. The dark knowledge of a pre-trained language teacher model could act as a regularization process, helping to calibrate the student model's confidence in its predictions.**
- 2. Unlike the baseline mixup method that focuses more on the class-specific features, by incorporating training dynamics into the mixup process, we focus more on the instance-specific major features that lead to the more calibrated models.**
 - ✓ Hosseini and Caragea (2022) show that student models distilled from teacher models trained using their proposed mixup method not only achieve the best performance but also obtain the lowest expected calibration errors (ECEs) on both in-domain and out-of-domain test sets on emotion-related tasks.**